

PERFORMANCE WORK STATEMENT (PWS)

Title: Autonomous Food Production System – 6-Month Pilot Program with Optional Containerized Model

Location: Camp Walker, Area IV, South Korea

Solicitation Number:

Contract Type: Firm Fixed Price (FFP) under Simplified Acquisition Threshold (SAT)

NAICS Code:

PSC Code:

Reference Number:

Performance Period: 6 months from date of award, with an option at Month 4 for deployment of a containerized model

Government Points of Contact:**Contracting Officer (KO):**

Name: [Insert KO Name]

Email: [Insert KO Email]

Office: 19th Expeditionary Sustainment Command, Korea

Contracting Officer's Representative (COR):

Name: [Insert COR Name]

Email: [Insert COR Email]

Duty Location: Camp Walker, Korea

Food Program Technical POC:

Name: CW3 River Mitchell

Title: Food Advisor, 19th ESC

Email: [Insert your email]

Phone: [Insert phone]

Vendor Point of Contact:

[Insert company name]

POC Name: [Insert]

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SECTION 1: Scope

Purpose:

This Performance Work Statement (PWS) defines the requirements for a 6-month pilot deployment of an automated food production system at Camp Walker, South Korea, with an option at Month 4 for deployment of a containerized model in a tactical or contingency location. This pilot will test modernized feeding capabilities to enhance 24/7 meal access, improve Soldier nutrition, and enable reallocation of culinary labor in support of Force Design 2030.

Objectives:

- Provide 24/7 access to high-quality, H2F-compliant meals.
- Increase DFAC utilization and reduce out-of-pocket spending.
- Reduce dependency on 92G Soldiers for garrison support by shifting capacity to tactical operations.
- Demonstrate technical, nutritional, and cost performance in alignment with the FY26 USFK Posture Statement and Holistic Health and Fitness (H2F) goals.

SECTION 2: Applicable Documents

The following regulations, standards, and guidance documents are informative references to this pilot and shall be followed unless otherwise stated. The contractor is responsible for staying current with any updates during the period of performance.

Nutrition and Food Program Standards:

- AR 30-22 – Army Food Program
- DA PAM 30-22 – Operating Procedures for the Army Food Program
- AR 40-25 – Nutrition Standards and Education
- FM 7-22 – Holistic Health and Fitness
- Go for Green® Guidelines – DoD Performance Nutrition Campaign (latest edition)

Sanitation and Safety Standards:

- TB MED 530 (Tri Service Food Code) – Occupational and Environmental Health: Food Sanitation
- NSF/ANSI Standard 4 (2023) – Commercial Food Equipment Sanitation
- HACCP – Hazard Analysis and Critical Control Points Guidelines
- FDA Food Code (2022) – Applicable to Machine food handling and labeling
- UFC 3-600-01 – Fire Protection Engineering for Facilities
- NFPA 96 (2024) – Ventilation and Grease Vapor Control
- NFPA 70 (NEC) – Electrical Installation Standards
- 29 CFR 1910.1200 – OSHA Hazard Communication Standard

Other Governing Guidance:

- FY26 USFK Posture Statement – Strategic priorities for sustainment and food access
- Army Futures Command Modernization Objectives – Innovation-aligned transformation targets
- Transformation in Contact (TIC) Implementation Framework – Sustainment modernization doctrine
- DLA Troop Support Prime Vendor Catalog – US Foods (Korea-based)

Cyber and Electrical Systems Standards

- NIST SP 800-171 – **Cybersecurity for Controlled Unclassified Information**
- DISA STIG (IAVM) – **DoD Information Assurance & Network Security Protocols**

SECTION 3: Requirements

The contractor shall provide a turnkey automated food production system to support a 6-month pilot program at Camp Walker, South Korea, with an option at Month 4 for deployment of a containerized model. The system must be fully operational, compliant with Army nutrition and sanitation standards, and capable of supporting 24/7 self-service meal access. The system shall follow all safety and electrical configuration requirements covered in this section.

3.1 System Installation & Site Preparation

- Install one fixed autonomous kitchen system at Camp Walker, within or adjacent to a designated DFAC or barracks.
- Conduct a joint pre-deployment site survey with 19th ESC and DPW to verify:
 - Electrical and water hookups
 - Ventilation and drainage
 - Network connectivity (Wi-Fi/cellular fallback)
 - Spatial access for setup and maintenance
- Vendor is not responsible for additional costs covering infrastructure compatibility.
- Bilingual And Korean language for both operations and ordering will be provided.
- The vendor shall provide detailed schematics showing electrical, water, and drainage requirements specific to the proposed autonomous kitchen system. These schematics shall be matched to the as-built drawings of the designated installation site to verify infrastructure compatibility, required utility drops, and space/pavement needs for turnkey operations.

3.2 Meal Production Requirements

- The Army will procure and provide all food ingredients through its existing Prime Vendor contract and catalog; the contractor is not responsible for purchasing food.
- During menu development, the contractor and Government will work collaboratively to design cyclic menus aligned to the current fiscal year BDFA for South Korea (FY2025 BDFA: \$25.84/day; Breakfast \$6.46, Lunch \$10.34, Dinner \$9.04). Menu planning shall use ingredient prices from the Government-provided Prime Vendor catalog.
- The Government Dietitian will review proposed menus during ramp-up to confirm nutritional compliance (AR 40-25; H2F guidance) and that average plate costs per meal type remain within BDFA planning rates unless a COR-approved exception is documented.

Note: The Government will provide the applicable BDFA rates for the current fiscal year at the time of menu development; the contractor will use only the most current rates provided

3.3 Digital Access & User Interface

- The contractor shall provide a touchscreen kiosk interface from Month 1 to support the 6-month pilot. The interface must enable:
 - Menu item selection
 - Order confirmation

- QR code-based transaction checkout
- The contractor shall provide a pick up screen interface from Month 1 to support the 6-month pilot. The interface must enable:
 - Real-time order monitoring
 - Locker box opening
- The contractor shall provide a touch screen Robot Control interface from Month 1 to support the 6-month pilot. The interface must enable:
 - Ingredient and product management
 - Menu creation and inventory tracking
 - System monitoring
 - POS capability for guest pay-by-meal (contractors, TDY, civilians) via one time use QR printer
- **Optional Future Capabilities (Not Required for Pilot):**
 - The contractor shall propose a plan for implementing a pre-order mobile application (Android/iOS) to enable pre-orders, scheduled pickups, and personalized meal suggestions.

3.4 Food Safety & Sanitation

- System shall meet:
 - TB MED 530, and HACCP protocols
 - Maintain safe temps ($\leq 41^{\circ}\text{F}$ cold for cold compartment)
- Include:
 - Dashboard visibility of sanitation history, temperature logs, DD Form 2973

NSF/ANSI 4 Compliance (Food Safety)

- All food-contact and food-zone surfaces must meet NSF/ANSI Standard 4 for commercial foodservice equipment.

Vendors must provide either full NSF/ANSI 4 certification, or

- A list of non-certified components with proposed mitigation strategies or timeline to obtain certification.
- This ensures food safety, hygiene, and cleanability standards are upheld during and after the pilot.

3.5 System Uptime & Support

- Maintain $\geq 80\%$ system availability (excluding scheduled maintenance, daily cleaning and ingredients refill).
- Enable remote diagnostics, firmware updates, real time telemetry and fault alerts.
- Provide:

- 12-hour response for software issues
- 48-hour response for on-site hardware support
- Forward spare parts kit stocked locally

3.6 Entitlement Tracking & Reporting

- From Month 1, the interface must authenticate meal entitlement via QR codes refreshed weekly. The system shall:
 - Export daily transaction logs tagged by meal period (B/L/D), entitlement type (meal card/guest), and anonymized ID.
 - Integrate data into AFMIS-compatible formats (CSV or equivalent) for COR, Dietitian, and DFAC use.
 - Support a sample size of 150–200 Soldiers KATUSAs and Coalition Forces (ROKA), selected by the COR in coordination with the DFAC Manager, to verify meal entitlement during the first two weeks of the pilot (Month 1). The sample shall include a representative mix of meal card holders (SIK/EUM eligible personnel). Verification results shall be documented in Deliverable 2 (Month 3 Initial Performance Report).
 - Provide a secure interface for the DFAC Manager to upload a weekly roster of authorized meal card holders in .CSV format, containing anonymized identifiers. The system shall:
 - Refresh QR codes weekly to reflect the updated roster.
 - Support manual overrides for ad-hoc additions/removals (e.g., TDY personnel).

3.7 – Fire Safety and Ventilation Requirements

To ensure safe and code-compliant deployment of autonomous food production systems within a U.S. Army garrison facility, the contractor shall meet the following fire protection, life safety, and ventilation requirements, in accordance with UFC 3-600-01, NFPA 96, AR 420-1, NEC, and all applicable DoD policies.

Ventilation System Standards (NFPA 96 / UFC 3-600-01 Compliance)

Cooking systems that produce grease-laden vapors must include a ventilation system compliant with NFPA 96 and UFC 3-600-01. Range hood fire extinguishing systems must be wet chemical or water-assisted chemical, designed and installed in accordance with NFPA 96.

If the system does not include a traditional Type I exhaust hood:

- The contractor shall submit independent engineering documentation or third-party testing demonstrating equivalent performance in removing grease, smoke, and vapor.
- Ventilation system specifications must be submitted to the Garrison Fire Marshal for review.

- The contractor shall acknowledge that lack of a compliant exhaust hood may require a formal Risk Acceptance Memorandum (RAM) signed by the Garrison Commander.

Note: For the initial 6-month pilot, systems may operate without a traditional hood only if:

- Independent documentation verifies the system does not generate grease-laden vapors (e.g., ≤50g oil per meal, no deep fryers);
- The cooking chamber is enclosed and powered by induction or similar non-flame technology;
- Fire load is minimal and operation is unattended;
- A portable CO₂ extinguisher (minimum 6 kg) is staged in the same room.

Fire Suppression Requirements (UL 300 or Equivalent)

Systems with heating components must be protected by a fire suppression system meeting UL 300 or documented equivalent standards approved by the Fire Marshal.

If suppression is not integrated into the system:

- The contractor must document the heating mechanism, cooking classification, and risk level;
- A portable extinguisher shall be staged within the unit's footprint;
- A RAM must be initiated with DPW and the Fire Marshal.

Egress Compliance (UFC 3-600-01 / AR 420-1)

- Autonomous systems shall not obstruct or degrade required egress.
- If installed in a space with only one exit:
- The contractor must submit a site layout confirming unobstructed traffic flow;
- The unit must not require human supervision during operation;
- A Commander-level RAM may be required for installation in single-egress areas.

Cooking Classification and Grease Vapor Designation

The contractor shall provide documentation assessing:

- Cooking method (e.g., induction, enclosed convection);
- Maximum oil use per dish or per hour;
- Grease containment features or filters.

If minimal or no grease vapors are produced, documentation shall support a low-risk classification for Fire Marshal concurrence during the pilot.

Emergency Power-Off (EPO) Capability

All systems must include an Emergency Power-Off (EPO) switch that:

- Is visibly mounted and labeled;
- Cuts power to all heating components in ≤ 5 seconds;
- Is resettable only by authorized personnel;
- Is tested during installation (deliver schematic and test log).

Post-Pilot Compliance Milestone Plan

By Day 150 of the pilot period, the contractor shall submit a milestone plan outlining:

- Modifications or additions IAW UL 300, NFPA 96, UFC 3-600-01, and NEC compliance;
- Timeline for NRTL-certified testing (e.g., UL 1978, ISO 9705);
- Equipment upgrades (if needed) for suppression, ventilation, or egress support.

The plan shall be submitted to the Contracting Officer and Garrison Fire Marshal no later than 30 days before the pilot concludes.

3.8 – Electrical Classification and Installation Standards

To ensure plug-and-play compatibility, electrical safety, and compliance with U.S. Army infrastructure and inspection protocols, all autonomous cooking systems deployed under this contract shall meet the following electrical and certification requirements, aligned with NFPA 70 (NEC), UFC 3-520-01, and AR 420-1.

Power and Compatibility Requirements

The system must operate at 208V, 3-phase, 60Hz without the need for external transformers or frequency converters.

All internal subsystems (e.g., motors, induction modules, controllers) must be validated for continuous 60Hz U.S. operation, without risk of overheating, degradation, or instability. For the initial 6-month pilot, systems may operate in their current configuration if documentation is provided confirming:

- Compatibility with U.S. Army 60Hz/208V power supply;
- No adverse effects on performance, reliability, or safety due to voltage/frequency differences;
- Electrical load does not exceed available breaker and circuit protection limits at the pilot site.

Plug or Disconnect Standard

Each system shall include either:

- A NEMA L21-30R (30A, 3-phase) or equivalent UL-listed weatherproof plug, or
- A UL-listed fused disconnect switch with integrated overcurrent and ground fault protection.

Circuit Protection and Monitoring

The system must incorporate:

- Overcurrent protection rated for $\geq 63A$, 60Hz service;
- A residual current device (RCD) or GFCI rated at 30 mA trip threshold;
- A visual ground fault indicator, such as an exterior LED or fault light.

Documentation Requirements (Pilot Phase)

Within 30 calendar days of pilot start, the contractor shall submit:

- Full electrical specifications and current draw/load profile;
- Plug/disconnect wiring diagrams and circuit schematics;
- Motor and subsystem frequency compatibility documentation.
- These submissions will be reviewed by the COR, DPW Electrical, and Garrison Safety/Fire Protection officials prior to system energization.

Post-Pilot Certification and Compliance Milestone Plan

By Day 150 of the pilot period, the contractor shall submit a follow on milestone plan including:

- A timeline for NRTL certification of all internal electrical components (UL, ETL, CSA, etc.)
- Any field evaluation testing (FET) or NEC 110.3-compliant verification required for components not currently certified;
- Identification of any subcomponents lacking NRTL listing, with proposed mitigation or test strategy.

The plan shall be submitted no later than 30 days before pilot conclusion to the Contracting Officer and COR.

3.9 – Containerized Autonomous Kitchen Requirements (Prototype Configuration)

If KPIs are met during the initial 3-month technical evaluation, the Government may exercise an option at Month 4 to deploy a prototype containerized version of the autonomous kitchen system at a tactical or contingency location within Area IV. This deployment is part of a limited-duration prototype test (MAR–SEP 2026) and is not intended to represent a final production configuration.

Execution of this option is contingent upon FY26 funding availability.

The vendor shall include optional pricing for this prototype deployment in the original proposal, to include:

- Door-to-door delivery (FOB Destination)
- Site installation and integration support, including physical setup, utility connections, and system activation
- A second site survey at the new deployment location to confirm infrastructure compatibility
- This prototype container shall be fielded for test and evaluation only and may operate under temporary compliance measures where full code compliance is not immediately feasible. Full compliance plans must be submitted for post-prototype scaling.

Container Prototype Requirements

- Structural Design and Safety
- ISO 1496-1 and MIL-STD-209K compliant container
- Seismic anchoring per UFC 3-301-01 (Zone 4)
- Two egress doors (opposing sides), each with panic bars
- Exterior: CARC green or coyote brown (FS 595B) with $\leq 15\%$ reflectivity or Government-approved branding
- Wind resistance: ≥ 85 mph with tie-down points compatible with Military Ground Anchoring System (MGS)

Electrical Power Integration (Prototype)

- 208V 3-phase 60Hz compatibility or local generator input
- Weatherproof NEMA L21-30R plug with hinged protective cover

Internal Power Generation (Post-Prototype)

By Day 60 of the prototype period, vendor shall submit a container-integrated power generation plan addressing:

- Power output rating (kW) and peak load capacity
- Fuel type and runtime per refill or charge
- Integration method with container power input and basic safety controls

Water and Plumbing

- $\frac{3}{4}$ " camlock potable water inlet with NSF 61-certified backflow preventer
- All internal tubing must be 304L stainless or FDA-grade silicone

Gray Water Collection and Discharge (Prototype Configuration) Discharge-ready connections shall include:

- Gravity-fed drain with manual shutoff

- 1" pumped outlet with camlock or threaded hose fitting

Internal Holding Tank Plan (Post-Prototype) Vendor shall present a plan including:

- Optical sensors (25% / 50% / 75% / 100%)
- NSF 51 antimicrobial coating
- OSHA-compliant "NON-POTABLE WASTEWATER" label with hazard pictograms

Note: No vendor-operated disposal is required. A Gray Water Connection Plan must be submitted 30 days prior to deployment, specifying compatible hookup methods to base infrastructure.

Ventilation (Prototype Configuration)

During the prototype period, the unit shall include a basic commercial exhaust solution capable of supporting eight induction burners, including:

- ≥600 CFM fan
- Carbon and HEPA filtration
- Exterior venting away from intakes and personnel areas
- Manual shutoff switch for all cooking appliances

Ventilation Compliance Plan by Day 60 of prototype operation, vendor shall submit:

- Timeline and pathway for full NFPA 96 and UFC 3-410-01 compliance
- UL 762 fan integration and grease duct specs
- Drawings and installation requirements for permanent Type I hood

Fire Protection and Life Safety

- UL 300 fire suppression system required (no exceptions)
- EPO switches at each exterior entrance and primary cooking zone
- Integrated CO and H₂S detection with ≥85 dB strobe/horn
- Class K fire extinguisher (10-lb) mounted at main entrance

HVAC & Environmental Monitoring

During the initial 6-month prototype evaluation period, the Government will accept systems operating with baseline HVAC and air purification features (e.g., plasma, HEPA, or equivalent), provided the following conditions are met:

- Milestone Plan Requirement (Within 30 Days of Award): Vendor shall submit a written milestone plan describing the pathway, timeline, and intended specifications to achieve:
- Integration of MERV 14 and carbon filtration meeting or exceeding UL 1978 standards
- Addition of N+1 cooling redundancy or documented mitigation strategy for critical temperature regulation

- Implementation of 24/7 environmental monitoring with remote alert capability for temperature, humidity, and airflow anomalies

Prototype Data Collection: Vendor shall monitor and record internal temperature and humidity throughout the prototype period to establish baseline performance.

Post-Prototype: Full HVAC and environmental monitoring compliance will be required for any transition to hardened, containerized, or multi-site operational deployments.

Deliverables – Container Prototype Phase

The vendor shall submit the following no later than 30 calendar days prior to prototype installation. For any item not available by this date, the vendor must provide a written milestone compliance plan detailing the timeline, third-party validation approach, and risk mitigation strategy:

- UL 300 Certification Report
- Environmental Tolerance Summary and Preliminary MIL-STD-810H Compliance Roadmap
- Ventilation Compliance Plan (Phase II)
- Gray Water Connection Plan
- On-Board Power Generation Plan (Phase II)
- Explosion-Proof Utility Seal Diagrams
- Container Specification Sheet with Interior and Exterior Layout
- Cybersecurity Checklist (aligned with DISA STIG and NIST SP 800-171 standards)
- Emergency Power Off (EPO) Shutoff Test Results (≤ 5 seconds for heating/cooking system)
- Wind Load Certification Signed by a Licensed Professional Engineer (PE)

Note: The Government reserves the right to request demonstration, third-party documentation, or on-site verification of any safety-critical system or certification as a condition of prototype acceptance.

3.11 – COUNTRY OF ORIGIN, NDAA SECTION 889, AND TAA COMPLIANCE

The Contractor shall ensure all components of the autonomous kitchen system comply with U.S. Federal acquisition laws, including the National Defense Authorization Act (NDAA), Trade Agreements Act (TAA), and Defense Federal Acquisition Regulation Supplement (DFARS). Noncompliance constitutes grounds for termination.

NDAA Section 889 Prohibition

- The system shall not include telecommunications equipment, video surveillance hardware, or IoT sensors from covered entities listed under 48 CFR §52.204-24 (NDAA §889(a)(1)(B)), including Huawei, ZTE, Hytera, Hikvision, or Dahua.
- A signed compliance certification (FAR 52.204-26) shall be submitted prior to installation.

TAA Compliance and DFARS Sourcing

- All hardware, software, and critical subcomponents (e.g., controllers, sensors, networking equipment) must be:
- Manufactured or substantially transformed in a TAA-designated country (e.g., U.S., South Korea, Germany, UK), or
- Covered under DFARS 225 exemptions applicable to COCOM operations.
- Non-TAA components (e.g., passive resistors or plastics) shall not exceed 20% of total contract value and may not include sensitive systems. These must be disclosed in a Bill of Materials (BoM) with the country of origin, function, and final assembly location.

Final Assembly and Testing

- Final integration, software loading, and functional testing shall occur in a TAA-compliant country (e.g., South Korea, Germany, or the United States).
- The contractor shall provide a Certificate of Final Assembly specifying the location of system integration.

Required Documentation

The following documentation shall be submitted to the COR no later than 30 calendar days prior to shipment:

- Bill of Materials with Country of Origin (per DFARS 252.225-7000)
- NDAA §889 Compliance Certificate
- TAA Compliance Statement (DFARS 252.225-7021)
- Certificate of Final Assembly Location

Noncompliance Consequences

Failure to certify or verify compliance will:

- Delay milestone payments under Section 5
- Trigger Government inspection and audit rights
- Permit termination for default under Section 5 provisions

3.12 Contingency and Exercise Support

The Autonomous Kitchen System may be deployed and operated within the Republic of Korea and, when required to support exercises, contingency operations, or technical evaluations, may be transported and temporarily operated in other locations within the U.S. Indo-Pacific Command (INDOPACOM) Area of Responsibility (AOR). These locations may include, but are not limited to, Japan, the Philippines, Guam, and Thailand. Cross-border deployments shall be coordinated in advance with the Contracting Officer and comply with applicable SOFA and host nation requirements.

The Contractor shall ensure that the system's modular components, operational controls, and support protocols are suitable for transport and operation at alternate locations outside of Korea during the pilot period, if directed. The Contractor shall be capable of supporting remote troubleshooting and virtual assistance for deployed operations. Physical support at non-Korean locations, if required, shall be coordinated through the Contracting Officer and may require additional authorization.

3.13 Scalability and Future Production Capability

The vendor shall demonstrate the autonomous kitchen's ability to scale beyond the initial pilot. This includes outlining future production configurations (e.g., containerized units, increased throughput models, or modular stackability) and providing technical options to support expansion across multiple sites. The system design must allow for scalable production volume, modular unit growth, and interoperability with existing Army food service infrastructure. Proposed scalability options will be evaluated for alignment with future field feeding and garrison sustainment objectives.

SECTION 4: Evaluation and Quality Assurance Plan

This section defines the performance standards, monitoring responsibilities, and evaluation timeline for the 6-month pilot of the autonomous food production system at Camp Walker. The COR, DFAC Manager, and Dietitian will validate all data using AFMIS logs, vendor dashboards, and Soldier feedback.

4.1 Key Performance Indicators (KPIs)

KPI 1: Meals Served by Machine

- Target: 80 meals produced per hour
- Method: Vendor export
- Threshold: $\geq 90\%$ of target (72 meals/hour)

KPI 2: System Uptime

- Target: $\geq 80\%$ availability
- Method: System logs and COR validation
- Threshold: ≤ 144 hours downtime/month

KPI 3: Meal Prep Time

- Target: ≤ 8 minutes average from order to meal completion
- Method: Timestamp analysis from robot control center
- Threshold: $\geq 95\%$ of meals meet standard

KPI 4: H2F Menu Compliance

- Target: 100% alignment with H2F and AR 40-25
- Method: Monthly menu validation by dietitian
- Threshold: No unapproved substitution

KPI 5: Entitlement Verification Success Rate

- Target: $\geq 95\%$ of sampled Soldiers successfully authenticate using QR code
- Method: System logs and COR validation
- Threshold: $\geq 90\%$ success rate

4.2 Oversight Roles and Monitoring Responsibilities

- **COR:** Validates data, approves milestone payments, oversees KPI performance
- **DFAC Manager:** Supports AFMIS headcount uploads, site ops, usage and waste tracking, and provides weekly roster updates
- **Dietitian:** Reviews and approves monthly menus for compliance
- **Vendor:** Provides dashboard access, issue response, and KPI reporting

4.3 Evaluation Timeline

- **Month 1:**
 - System installation and baseline validation
 - Entitlement verification
 - Real time dashboard and export review
- **Months 2–4:**
 - Ongoing monitoring of KPIs
 - Mid-cycle corrections if performance drops
 - Decision point for containerized model deployment
- **Month 5:**
 - Formal KPI checkpoint
 - Final evaluation and full performance report
 - Basis for option year decision (IDIQ) or system off-ramp

4.4 Data Validation Sources

- Vendor dashboards and automated reports
- Soldier feedback surveys
- Manual reviews by COR, DFAC Manager, and Dietitian

4.5 Off-Ramp and Corrective Action Criteria

- If the contractor fails to meet two or more core KPIs by Month 3, or if significant operational or safety issues arise, the Government may elect to terminate the pilot. Any termination will follow the optional clause outlined in Section 5, and payments will be prorated based on completed milestones.

SECTION 5: Deliverables and Payment Milestones

The contractor shall complete the following phased deliverables over the 6-month performance period. Payments will be made upon government acceptance of each milestone. All deliverables must meet the quality standards outlined in Section 4 (QASP).

Expected Award date 01st September and product will be delivered NLT 30 September 2025 to start system installation and readiness.

Deliverable 1: System Installation and Readiness

- Due: Within 30 calendar days of contract award
- Completed site survey (Camp Walker)
- Fixed autonomous kitchen delivered and installed
- Network, power, and operational testing complete
- Staff trained on system operation
- Payment: 30% of contract value upon order placed

Deliverable 2: Initial Performance Report

- Due: End of Month 2
- Weekly meal volumes (Machine + DFAC)
- Prep time averages
- Waste rate tracking
- H2F menu validation
- System uptime report
- Entitlement verification results for 150–200 Soldiers
- Payment: No additional payment; used as a performance checkpoint

Deliverable 3: Month 4 KPI Milestone + Optional Containerized Model

- Due: End of Month 4
- Verification of KPIs from Section 4 (meals/week, uptime, waste, H2F, entitlement verification)
- Decision point for containerized model deployment (if KPIs met)
- If exercised, site survey and setup of containerized system
- Recommendations for expansion or follow-on effort, including a proposal for a pre-order mobile application (Android/iOS)
- Payment: 40% of contract value upon acceptance (includes containerized deployment if exercised)

Deliverable 4: Final Report and Closeout Brief

- Due: End of Month 6
- Full KPI performance summary
- Soldier feedback analytics

- Labor reallocation tracking
- Lessons learned and scalability potential
- Payment: Final 30% of contract value upon acceptance

Option Year Consideration Clause:

- By the end of Month 5, the Government will determine whether to initiate a follow-on requirement (e.g., Base Year + or permanent system retention). If pursued, the contractor shall support transition planning, data handoff, and continuity of operations. The contractor shall maintain all equipment in-place unless otherwise directed by the Contracting Officer.

Late or Incomplete Deliverables:

- Payment for any deliverable may be withheld until COR verifies satisfactory completion and data accuracy.
- If the government elects to terminate after Month 3 due to poor performance, payments will be prorated to cover completed deliverables only.

SECTION 6: Roles and Responsibilities

This section defines the key stakeholders and their respective responsibilities to ensure effective execution, oversight, and evaluation of the pilot program.

Contracting Officer (KO)

- Serves as the only individual with authority to modify the contract.
- Ensures compliance with FAR/DFARS provisions.
- Manages contract terminations, extensions, or award of follow-on actions.

Contracting Officer's Representative (COR)

- Serves as the government's primary technical liaison.
- Monitors contractor performance in accordance with the QASP.
- Verifies deliverables and authorizes milestone payments.
- Coordinates site access, compliance reviews, and government-furnished info.
- Validates entitlement tracking accuracy and meal card holder access.

DFAC Manager

- Assists with AFMIS uploads and reporting.
- Coordinates on-site operations, equipment support, and issue resolution.
- Reviews weekly data on meal volume, usage trends, and food safety compliance.
- Coordinates with the COR to select the sample size of Soldiers for entitlement verification and provides weekly roster updates for authorized meal card holders

Registered Dietitian (or designee)

- Reviews menus for H2F alignment and MDRI compliance.
- Approves meal tagging for nutrition labels and dietary goals.
- Provides feedback on Soldier performance nutrition needs.

Garrison DPW

- Provides coordination for facility access, power, safety, and water during installation.
- Assists with pre-deployment site surveys.
- Ensures compliance with safety, electrical, and infrastructure requirements.

Contractor

- Responsible for full system deployment, operation, training, and on-site maintenance.
- Provides real-time dashboard access and all required reports.
- Ensures system uptime, sanitation and safety compliance, and user support.
- Provides all disposables, cleaning supplies, and commercial connectivity.

- Coordinates with COR and DFAC manager for all on-site tasks.

SECTION 7: Security Requirements and Cyber Compliance

The contractor shall implement commercially reasonable cybersecurity, information assurance, and physical security measures to protect the integrity and data of the autonomous food production system when operating on commercial Wi-Fi or cellular networks. These measures shall be aligned with industry best practices and applicable DoD guidelines to the extent practicable during the pilot.

7.1. System Architecture and Network Security

- The system shall operate independently and shall not connect to NIPRNet or SIPRNet.
- The system shall use commercial Wi-Fi or cellular networks secured by firewall protections or equivalent commercial-grade safeguards.
- All data transmission shall utilize TLS 1.2 or higher, or a comparable encryption protocol, to protect in-transit data.
- Stored system configuration and operational data should be encrypted using AES-256 or equivalent, where technically feasible.

7.2. Cybersecurity Standards

- During the pilot, the contractor shall adopt basic cybersecurity controls such as user authentication, system hardening, and monitoring consistent with commercial best practices.
- Prior to the conclusion of the pilot, the contractor shall provide a Cybersecurity Compliance Milestone Plan identifying its pathway to aligning with:
 - NIST SP 800-171 – Controlled Unclassified Information (CUI) protections
 - DFARS 252.204-7012 – Safeguarding Covered Defense Information
 - DoDI 8520.03 – Identity Authentication
 - DISA STIG/IAVM protocols – DoD Information Assurance standards

7.3. Data Handling and Privacy

- No Personally Identifiable Information (PII) or Controlled Unclassified Information (CUI) shall be stored on the system.
- Soldier interactions with the system shall be anonymized (e.g., via QR code or unique non-PII identifier).
- Meal transaction logs shall be exportable in an anonymized format, including entitlement status, meal type, and timestamp, to support operational reporting.

7.4. Access Control and Permissions

- The system shall support role-based access controls (e.g., admin, technician, dietitian, COR) to restrict access to authorized users.

- User accounts shall require password-protected logins, with activity tracking where feasible to ensure accountability.

7.5. Incident Reporting and Response

- Any suspected cyber incident shall be reported to the Contracting Officer's Representative (COR) within 48 hours or as soon as practicable.
- A summary report shall be submitted within 72 hours, including:
 - Date/time of event
 - Description of the incident
 - Steps taken to address the issue
 - Recommendations for preventing recurrence, if applicable

7.6. Cybersecurity Plan of Action and Milestones (POA&M)

- Within 60 days of contract award, the contractor shall submit a POA&M that:
 - Identifies cybersecurity measures implemented and any gaps specific to commercial Wi-Fi operation
 - Provides a proposed timeline for addressing identified gaps
 - Outlines procedures for ongoing monitoring to maintain system security

SECTION 8: User Interface and Experience

8.1. Access and Authentication

- For Meal Card Holders (SIK/EUM/KATUSA Eligible):
 - The system shall support seamless access and authentication for authorized meal card holders from Month 1 using the following platforms:
 - Touchscreen kiosk (on-site)
 - Meal entitlement verification shall use:
 - Non-PII QR codes tied to roster-synced eligibility (refreshed weekly for adding new users)

8.1.a Non-Meal Card Holders and Pay-by-Meal Capability

- A dedicated tablet and software interface shall be provided and installed at the participating dining facility where the autonomous kitchen is located. This capability enables non-meal card holders (e.g., civilians, contractors, TDY personnel) to access meals under standard Army procedures:
 - Guests will pay for meals via the DFAC cashier at standard meal rates
 - The cashier shall:
 - Log the transaction in Army POS system as a paid meal
 - Cashier provides guest QR code that enables them to select dishes at the robot
 - The system will:
 - Generate a secure, single-use QR code tied to the purchased meal(s)
 - Deliver the QR code via printed receipt or digital format (text/email), enabling either:
 - One-time robot access
 - This process ensures:
 - Single entitlement logic: Meals are recorded once, at the AFMIS point of sale
 - No dual crediting: The vendor system validates QR codes for one-time use only
 - No added burden on DFAC staff: Cashier workflow aligns with existing service procedures

8.2. Functional Capabilities

- Provide interactive nutrition labels (calories, macros, allergens, dietary tags)
- Include hydration prompts and balanced plate recommendations based on meal context

8.3. Feedback Collection

- Interface shall support:
 - Star ratings (1–5)
 - Optional comments
 - Brief surveys for menu suggestions or system feedback
- Feedback must be:
 - Stored securely and anonymously
 - Exportable to the COR via a monthly snapshot

8.4. Locker Management Logic

- System must show real-time locker capacity
- Auto-pause ordering when lockers are full
- Allow operators (with verified credentials) to manually clear expired lockers

SECTION 9: Reporting Requirements

The contractor shall provide real-time reporting and data export capabilities through a secure, cloud-based dashboard accessible to authorized government personnel.

9.1. AFMIS-Compatible Meal Reporting

- All meal transactions must be:
 - Timestamped
 - Tagged by entitlement type (meal card holder, guest pay)
 - Categorized by meal period (breakfast, lunch, dinner)
- Data must be exportable in AFMIS-compatible formats (.CSV or equivalent) for daily input and archiving.
- Reports must reset daily at 0000

9.2. Oversight Dashboard (COR, Dietitian, DFAC Manager)

- Must provide real-time visualizations of:
 - Meals served by time, day, and entitlement type
 - Peak demand trends (e.g., post-PT, weekends)
 - Real time inventory usage and active monitoring of usage
- Dashboard must allow:
 - Automated daily/weekly report emails (if requested)
 - Visual task checklists (cleaning logs, DD form 2973)

9.3. Inventory and Forecasting Dashboard

- Must display:
 - Ingredients in stock (SKU-level)
 - Usage trends and restock alerts
 - Food cost tracking tied to BDFA rates and prime vendor catalog

9.4. Food Safety Dashboard

- Must show live temperature monitoring of:
 - Refrigeration zones and include alerts for out-of-range temps

9.5. Production and Staff View

- Separate tablet or login-based interface for system operators (if needed)
- Display:
 - Daily prep quantities per menu item
 - Ingredient silo levels with traffic light system (Green = OK, Red = Low)
 - Daily cleaning task status
 - Task management for daily dish temperature checks

- Auto notifications when prep or sanitation thresholds are unmet

SECTION 10: Follow-On Acquisition Planning

10.1 Intent for Future Scalability This 6-month Technical Evaluation is structured to inform future acquisition strategies that may include a scalable, modular Indefinite Delivery/Indefinite Quantity (IDIQ) contract to support broader deployment of autonomous meal production systems across U.S. Army installations in Korea and other Indo-Pacific locations.

The Government will use validated performance metrics from this TE—including system uptime, meal output, user satisfaction, and nutrition compliance—to define potential scope, CLIN structure, and performance standards for any follow-on contracting action.

10.2 Use of Pilot Data for Acquisition Planning All performance data collected during the TE (including deliverables, logs, Soldier feedback, and KPI results) shall be considered Government-furnished data. This data may be used to inform:

- Future performance-based requirements
- Competitive evaluation criteria
- Market research and vendor pool development

10.3 No Implicit Guarantee of Follow-On Award Participation in this TE does not guarantee the award of a follow-on contract to the vendor. Any future acquisition, including a potential IDIQ, shall be subject to applicable Federal Acquisition Regulation provisions and competed as required unless a Justification and Approval (J&A) is executed by the Contracting Officer.

10.4 IDIQ Feasibility Assessment By the end of Month 5, the Government will assess whether to initiate planning for an IDIQ contracting vehicle based on the technical performance, operational benefit, and cost-effectiveness demonstrated by the vendor. If pursued, the contractor may be requested to assist with transition planning, market research input, and documentation of lessons learned.

SECTION 11: Conclusion and Government Coordination

This Performance Work Statement defines the objectives, roles, and expectations for a 6-month pilot deployment of an automated food production system at Camp Walker, South Korea, with the potential for a containerized model deployment and long-term adoption.

The pilot directly supports Army modernization priorities outlined in the FY26 USFK Posture Statement, Holistic Health and Fitness (H2F) implementation, and Force Design 2030 workforce realignment objectives.

11.1. Government Oversight and Coordination

To ensure pilot success, the Government will coordinate with the contractor through:

- Site planning and infrastructure prep in collaboration with DPW and 19th ESC.
- Monthly performance reviews led by the COR and supported by DFAC staff and dietitians.
- Live issue resolution, data validation, and system usage monitoring via integrated dashboards.
- KPI assessment checkpoints, especially at Month 3 and Month 6, to inform performance-based decisions.

11.2. Transition Planning and Continuity

- By Month 5, the Government will determine whether to:
 - Extend operations under follow-on procurement
 - Expand deployment scope (e.g., Base Year + or additional sites)
 - Off-ramp and close out the pilot with full equipment recovery
- The contractor shall:
 - Maintain system availability through contract end
 - Support data handoff, continuity planning, and final reporting
 - Remove or retain system components only as directed by the KO

This pilot establishes a scalable, Soldier-centered food access model aligned with strategic Army sustainment and readiness goals. Its success will inform decisions on wider implementation across the Indo-Pacific and garrison DFAC networks.

Annex 1: Abbreviations and Terms

- AR – Army Regulation
- AFMIS – Army Food Management Information System
- BDFA – Basic Daily Food Allowance
- CUI – Controlled Unclassified Information
- DFAC – Dining Facility
- DLA – Defense Logistics Agency
- DPW – Directorate of Public Works
- ESC – Expeditionary Sustainment Command
- FM – Field Manual
- H2F – Holistic Health and Fitness
- KO – Contracting Officer
- KPI – Key Performance Indicator
- NIST – National Institute of Standards and Technology
- NSF – National Sanitation Foundation
- PWS – Performance Work Statement
- QASP – Quality Assurance Surveillance Plan
- QR Code – Quick Response Code (used for non-PII meal entitlement authentication)
- SAT – Simplified Acquisition Threshold
- SIK – Subsistence-in-Kind (meal entitlement)
- SPV – Subsistence Prime Vendor
- TB MED 530 – Technical Bulletin Medical 530: Food Sanitation
- TIC – Transformation in Contact
- USFK – United States Forces Korea
- 92G – Culinary Specialist Military Occupational Specialty